ADAPTIVE STAIRCASE EXPERIMENT

PSY310 LAB PSYCHOLOGY

LAB REPORT

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GitHub Link:
https://github.com/diyas8-ui/PSY310_LAB/upload/Adaptive-staircase

Introduction:

An alternative method is an adaptive “staircase” procedure, an algorithm that actively adjusts the stimuli on-line in response to the subject’s performance [1]. The procedure adjusts according to pre-set rules and terminates when the stimuli approximate the subject’s threshold, making it a much more efficient method at the cost of lower data availability [2]. Due to this efficiency, staircase procedures are ideally suited for experiments targeting low perceptibility stimuli, and the algorithm logic has been described in depth previously [3], [4] and several variants have been developed (e.g. [5], [6]; for reviews see [7] or [8]). (Hariston & Maldjian, 2009)
The present submitted experiment is a task of orientation discrimination and to calculate its threshold for sinusoidal gratings using the adaptive staircase procedure.

Method:

Participant

The participant is a 19 year old female student of Ahmedabad University. The participant was made aware of the objective of the trial, that was to determine the direction of the tilt, moreover the participants consent was taken prior to the experiment, and no demographic information was recorded.

Materials

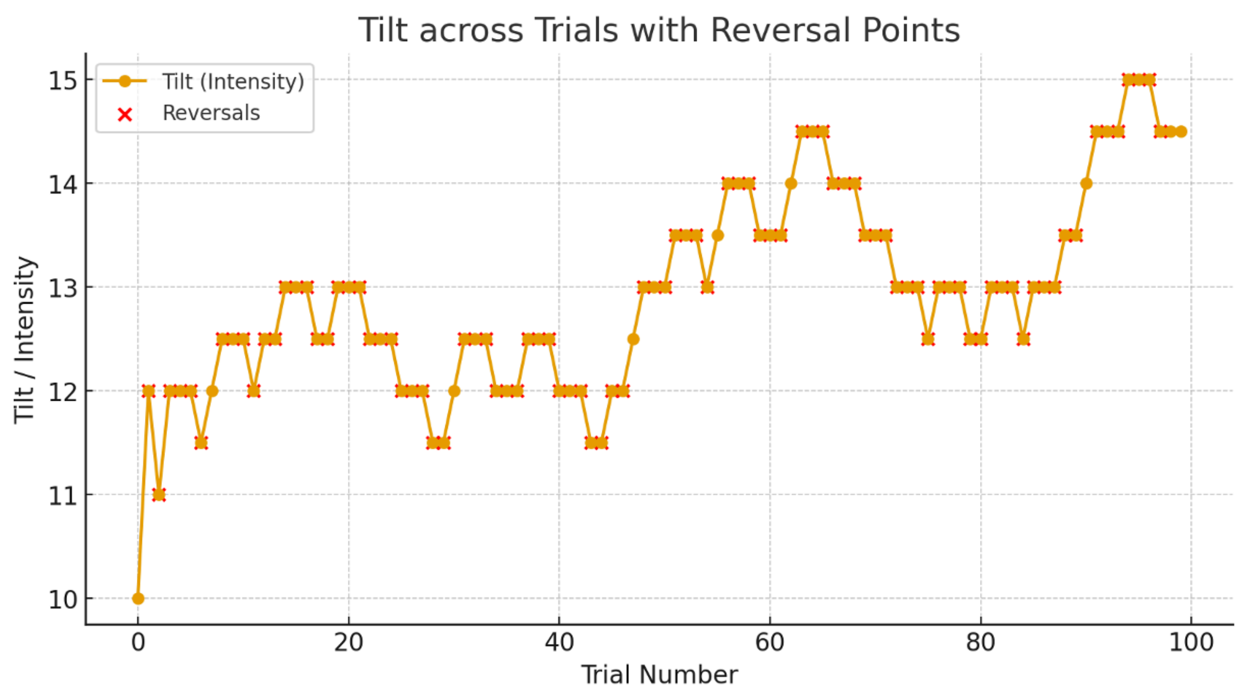
The experiment was conducted on an Apple Macbook M1 Air 2020 13” inch. The experiment was conducted through a software called Psychopy (2025.1.1), and all the necessary information was recorded through the same. A sinusoidal grating with a gaussian mask was presented as the stimulus. It was displayed on a plain grey background with a cross as the fixation.

Procedure

The experiment began with a white fixation cross for 100 ms, followed by a grating stimulus which tilted in either right or left direction randomly for 500ms. The right and left arrow keys were used to express responses by the participant during the experiment. A one up, three down rule was followed where in a total of three correct responses decreased the tilt and just one incorrect response increased the tilt of the grate. The range of the intensity of the tilt varied from 1 degree to 10 degrees. All responses were directly stored and saved in an excel file through psychopy. These records were later used during the analysis stage of the experiment.

Result:

A total of 100 trials were conducted. The overall accuracy of the participant was 67%. Which signifies that the participant did a relatively good job by judging the tilt of the grating stimulus correctly. The mean response time was 2.0531.
The further analysis of the adaptive staircase reversal displayed a convergence at a threshold of 1 degree, which proves that the tilts of the grating stimulus of that magnitude could be easily interpreted. Below attached are two graphs. Figure 1:shows the tilt of the grating across trials.



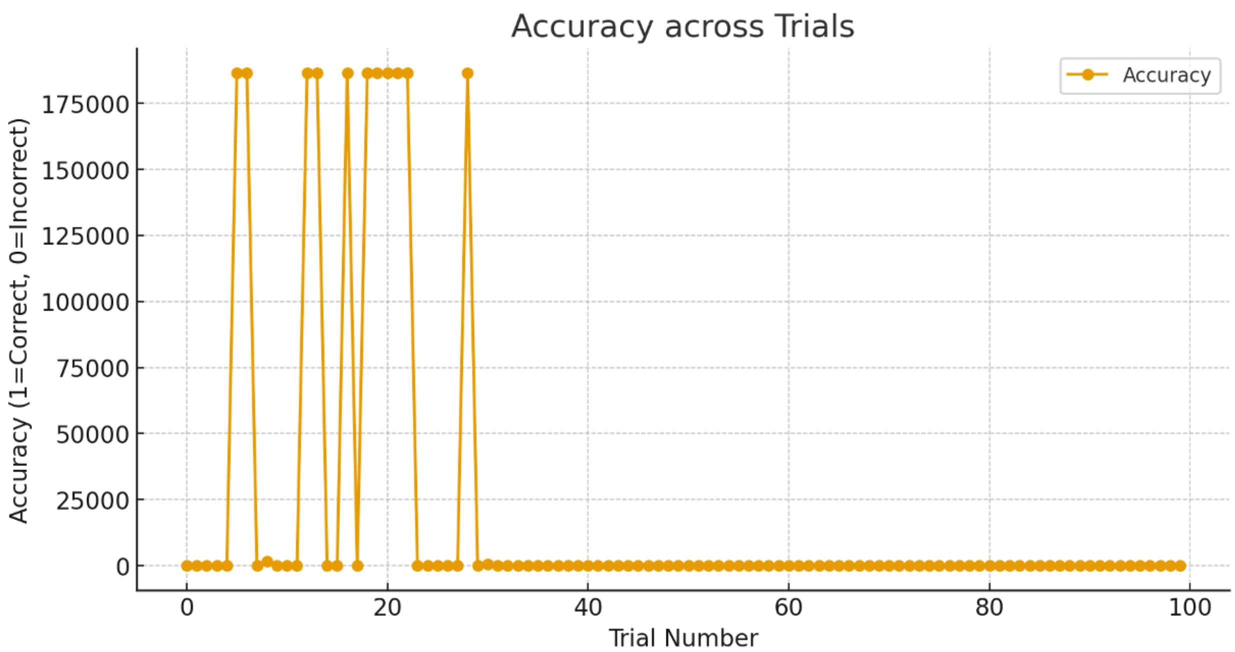


Figure 2: shows response accuracy across trials

Discussion:

There was a point where the procedure had converged at 1 degree, which shows the participant's awareness and understanding to discriminate between the tilt. The records and discussion were in sync with the previous research done supporting the efficiency of adaptive staircase in determining the threshold.

References:

https://www.sciencedirect.com/science/article/abs/pii/S016926070800206X